



L3-4 Judgment Exit Under Continuing Operation

**Chapter 3 — Operational Absorption:
When Unstable Conditions Are Sustained Through Behavioral Control**

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Project Background (Factual Summary)

Project year: 2018

- **Location:** Shanghai, China — within a food testing laboratory facility converted from an existing office building
- **Project type:** Renovation of two upper floors within an existing eight-storey office building into food testing laboratory space
- **Building profile:** A non-R&D office building retrofit project, where the original building structure and floor-to-floor height were not designed for laboratory use
- **Functional arrangement:** Integrated laboratory and administrative functions within a single building system
- **Project phase:** Progressed from schematic design into detailed design, followed by immediate construction execution
- **Stage of involvement:** Entered during the multidisciplinary coordination stage after primary system directions had already been established
- **Scope of work:** Interface coordination across laboratory planning, HVAC systems, MEP systems, and internal functional layout
- **Role positioning:** Operating at the intersection of laboratory planning and multidisciplinary coordination, responsible for system integration and interface alignment between client teams, consultants, and base-building discipline
- **System characteristics:** Laboratory functions inserted into an existing structural framework while simultaneously coordinating experimental workflow, personnel circulation, equipment layouts, and ceiling-level MEP organization
- **Supporting systems:** HVAC, plumbing, electrical systems, laboratory exhaust systems, control systems, and associated supporting infrastructure functioning as a unified operational framework, functioning as an interdependent operational framework.

Operational Narrative Overview

In this project, the laboratory systems were inserted into an existing office building structure that had never been designed for laboratory operation. As the project progressed, the available ceiling space gradually became saturated by HVAC ducts, exhaust systems, electrical services, sprinkler piping, and related supporting infrastructure. Under these constraints, portions of the exhaust ductwork adopted highly compressed geometries, while the physical distance between supply diffusers, exhaust terminals, and fume hoods was continuously reduced.

Although these conditions increasingly limited airflow organization and acoustic performance, the system itself remained operable throughout design development, commissioning, and final acceptance. The laboratory successfully completed the required testing procedures and entered formal operation without triggering a system-level redesign.

The instability only began to emerge after the laboratory entered real operational use. As personnel movement, simultaneous operation, and daily workflow density gradually returned to normal conditions, localized airflow disturbances near the fume hoods began producing continuous fluctuations in face velocity readings and repeated alarm activation.

However, by this stage, the system no longer possessed the physical capacity for structural correction. Ceiling space, diffuser locations, and overall airflow organization had already been fixed within the retrofit boundary conditions of the building.

As a result, the project did not return to redesign. Instead, the remaining instability was gradually absorbed through operational restrictions, behavioral controls, alarm suppression, and additional PPE requirements. What changed was not whether the laboratory could continue operating, but how the unresolved condition was ultimately sustained within operation itself.

1. Constraint Accepted as Operable Condition

As the project involved the retrofit of an existing office building, the original floor-to-floor height and ceiling void were never intended to accommodate laboratory systems. Once laboratory functions were inserted into the existing office structure, HVAC systems, exhaust ductwork, electrical services, sprinkler systems, and associated supporting infrastructure were all required to coexist within a highly limited ceiling space. As system density continued to increase, the ceiling zone gradually approached its physical limits.

Under these conditions, portions of the laboratory exhaust ductwork began adopting aspect ratios significantly beyond those typically found in conventional comfort-based systems. Several duct sections exceeded a 1:4 aspect ratio while remaining compressed within densely occupied ceiling areas. At the same time, the relative distances between supply air terminals, exhaust outlets, and laboratory fume hoods were progressively reduced due to the absence of remaining coordination space.

These conditions were not initially defined as system failure during the design stage. In retrofit laboratory projects, systems are often not developed around ideal conditions, but rather around the maximum level of coordination achievable within pre-existing structural constraints. In this project, although noise levels, ceiling congestion, and airflow organization had already begun to be affected by the limitations of the original building, the system was still capable of completing basic airflow balancing, equipment integration, face velocity testing, and subsequent commissioning procedures.

As design development and construction continued, these constraints never triggered a system-level rejection. The project continued advancing within the established direction and ultimately proceeded into commissioning and acceptance.

During the final acceptance stage, the overall laboratory systems successfully completed the required testing procedures. Although noise levels in certain areas had already exceeded typical comfort standards, the limitations imposed by the existing building conditions were clearly understood, and no meaningful adjustment capacity remained within the system. As a result, the condition was ultimately accepted by the owner and carried forward as part of the operational baseline of the project.

At this stage, the condition was not treated as failure.

It was not defined as an abnormal state requiring elimination, but rather accepted as an operational boundary condition within the constraints of the retrofit environment.

2. Stability Verified Only Under Controlled Conditions

During commissioning and acceptance, the laboratory fume hood systems generally maintained operating conditions that satisfied the required testing criteria.

The relevant testing procedures were conducted under relatively controlled environmental conditions. Personnel movement within the laboratory remained limited, equipment operation states were relatively stable, and sources of airflow disturbance were kept at a relatively low level. Under these conditions, the face velocity of the fume hoods remained within the required acceptance range without exhibiting continuous abnormal fluctuation.

At the same time, the primary focus of the project remained centered on whether the system could successfully operate within the physical boundaries of the existing building, rather than whether the system retained sufficient long-term operational stability margin. Under conditions where the ceiling void had already approached its physical limits and no additional coordination capacity remained available, the successful completion of commissioning and acceptance itself was regarded as a critical project milestone.



2. Stability Verified Only Under Controlled Conditions

At this stage, system stability itself was never fundamentally redefined.

What the testing procedures verified was that the system could maintain the required operational state under specific controlled conditions, rather than confirming that the system would remain continuously stable under all real operational scenarios.

As the laboratory gradually entered full daily operation, real operational conditions began progressively re-entering the system.

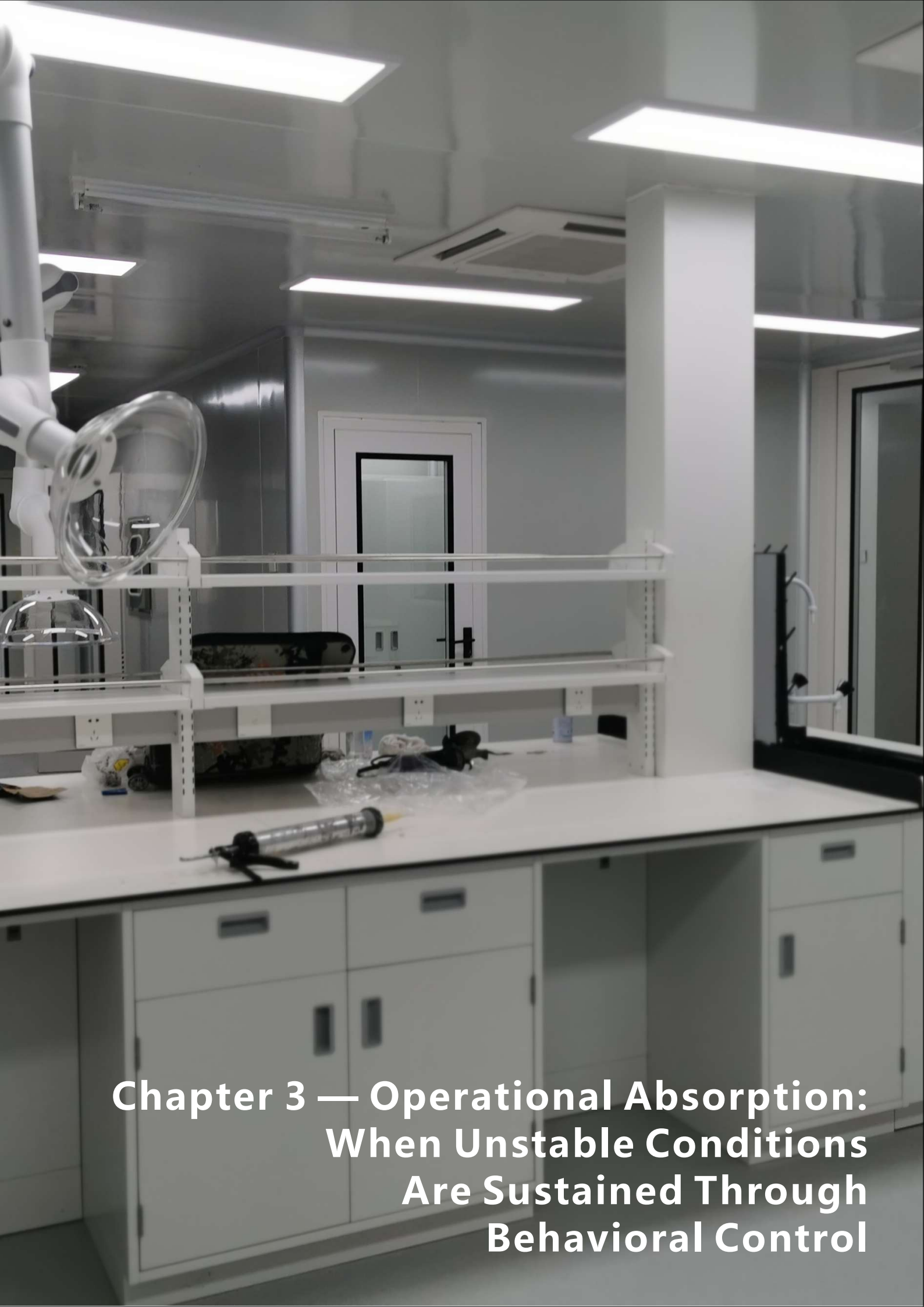
As personnel density increased, movement circulation intensified, multiple operators began working simultaneously, and normal logistical movement gradually returned to full operational frequency, conditions that had not appeared during commissioning slowly began to emerge. Due to the close proximity between supply air terminals and laboratory fume hoods, localized airflow disturbances generated by personnel movement in front of the hoods began continuously affecting face velocity stability.

At this stage, the system itself did not stop operating.

The HVAC systems, exhaust systems, and laboratory fume hoods all continued functioning according to their predefined operational logic. However, the stable condition that had previously existed under controlled testing environments gradually began deviating once exposed to real operational environments.

The resulting condition did not appear as system failure.

Instead, it appeared as a condition in which system stability had only been verified under controlled conditions.



Chapter 3 — Operational Absorption: When Unstable Conditions Are Sustained Through Behavioral Control



3. Behavioral Control Replacing System Correction

As the laboratory entered long-term operation, continuous face velocity fluctuations began repeatedly triggering system alarms.

Over time, the persistent and high-frequency alarm sounds gradually began affecting the working environment within the laboratory and interfering with daily experimental operations. At the same time, the existing building conditions had already been fully exhausted, leaving no remaining pathway for meaningful system-level correction. Ceiling space, air terminal locations, duct routing, and associated system relationships could no longer be substantially reorganized or adjusted. Under these conditions, the project did not re-enter a system redesign phase.

No new airflow organization strategy was established, nor were the supply and exhaust relationships fundamentally reconstructed. Instead, subsequent operation gradually shifted toward absorbing the remaining instability through operational discipline and behavioral restriction.

As a result, a series of new operational control measures gradually emerged within the laboratory environment.



3. Behavioral Control Replacing System Correction

Fixed standing boundaries were marked on the floor in front of the fume hoods, limiting the number of operators permitted within designated zones in order to reduce airflow disturbance generated by personnel movement. At the same time, the audible alarm function of the face velocity monitoring system was permanently disabled, while visual alarms were retained in order to reduce the continuous operational disruption caused by repeated alarm activation.

Additional PPE requirements were also introduced for personnel operating within the affected laboratory areas.

At this stage, the system itself did not regain operational stability.

What changed instead was that the operational organization of the laboratory gradually began assuming responsibility for the remaining instability that the system itself could no longer eliminate. Conditions that would traditionally require system-level correction were progressively transformed into long-term restrictions on personnel behavior, operational procedure, and workplace discipline.

The system itself did not become stable again. Instead, portions of the stability responsibility originally belonging to the system layer gradually became carried by the operation layer and by human behavior itself.

Chapter 3 — Conclusion

In this project, the system itself did not stop operating because of the limitations imposed by the existing building conditions.

Instead, the laboratory systems continued functioning within a highly compressed ceiling environment and successfully proceeded through commissioning and final acceptance. The resulting conditions were never defined as system failure during the design stage, but were instead accepted as operational boundary conditions within the constraints of the retrofit environment.

What later re-entered the system was not equipment failure itself, but the gradual deviation of a stable condition that had only been validated under controlled testing environments. As the laboratory entered long-term operation, personnel movement, localized airflow disturbance, and daily operational activities progressively revealed the limited stability margin retained within the system.

By this stage, however, the system no longer retained the ability to substantially reconfigure itself. Ceiling space, airflow relationships, and overall system organization had already become fixed within the existing structural framework, leaving no remaining capacity for meaningful system-level correction. As a result, subsequent operation did not restore stability through redesign, but instead gradually shifted toward absorbing the remaining instability through behavioral restriction, operational discipline, and additional operational protocols.

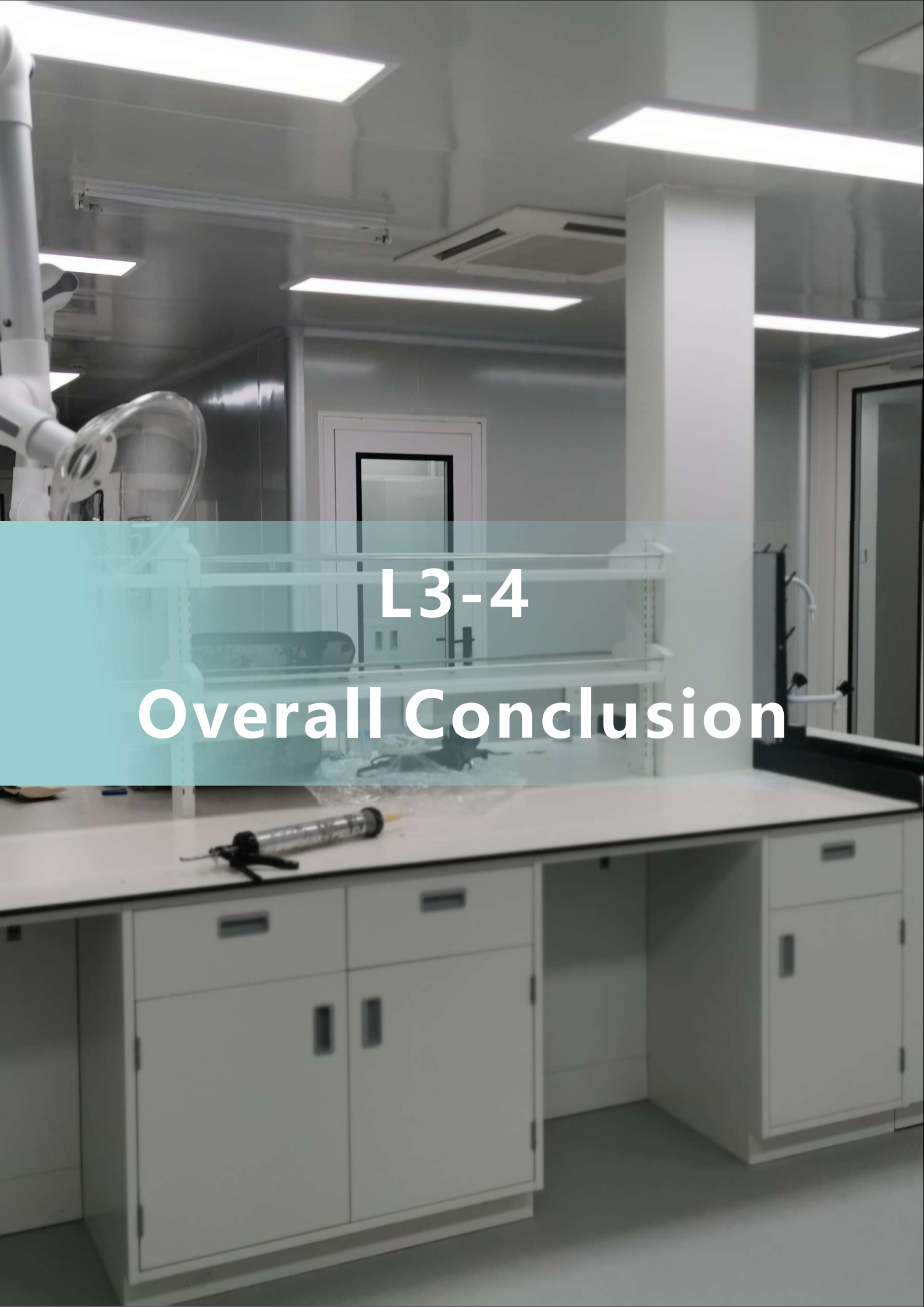
Throughout this process, the system itself never fully regained stability.

Instead, portions of the stability responsibility originally belonging to the system layer were progressively transferred to the operation layer and to human behavior itself.

Not all unresolved conditions lead directly to system failure.



**Some conditions gradually become institutionalized
as part of operation itself.**



L3-4

Overall Conclusion

L3-4

Overall Conclusion

Across all three chapters of L3-4, the system itself never fully stopped operating.

In Chapter 1, reagent storage requirements that could not be fully accommodated within the laboratory building were not resolved through internal spatial reorganization. Instead, the condition was displaced through the redefinition of the storage carrier itself, transferred into an external facility, and gradually stabilized as an accepted operational premise. Only after prolonged operational mismatch did the excluded condition re-enter the system through distributed buffer rooms and partial reconstruction within the laboratory interior.

In Chapter 2, the absence of thermal service during the transition season did not enter the system as a design-stage conflict. Because the interval remained outside the formally defined scope of system response, it gradually became stabilized through interpretation and coordination, eventually embedding itself into the operational structure of the project. Only during actual laboratory operation did the condition reappear as a persistent deviation between environmental control and real usage requirements.

In Chapter 3, airflow instability caused by spatial compression, constrained ceiling organization, and operational airflow disturbance did not trigger structural redesign after occupancy. Instead, the remaining instability was gradually absorbed through behavioral restriction, operational discipline, alarm suppression, and additional PPE requirements. The condition itself did not disappear; it was redistributed into the operation layer and continuously carried by human behavior.

Across all three cases, the system continued moving forward.

- Design development continued.
- Construction proceeded.
- Commissioning was completed.
- Operation remained possible.

L3-4 Judgment Exit Under Continuing
Operation
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• Planning • EHS Logic • Modularity • Commissioning

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Chapter 2 — Interpretation as Structure
When Undefined Conditions Are Stabilized
Resolution

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Overall Conclusion

What changed was not whether the system could continue operating, but where unresolved conditions were ultimately absorbed.

- Some conditions were transferred into time.
- Some were absorbed into interpretation.
- Some were redistributed into operation and human behavior.
- The system does not necessarily eliminate these conditions.

More often, it reorganizes the manner in which these conditions are allowed to continue existing.

In this sense, L3-4 does not document isolated design mistakes.

Instead, it records a recurring structural process repeatedly observed within complex project environments: conditions that fail to complete judgment during stages where structural correction remains possible do not truly leave the system. They continue forward through assumption, interpretation, buffering, or operational absorption, and eventually re-enter the project in another form.

System stability does not necessarily mean that all unresolved conditions have been eliminated.

Very often, stability simply means that the system has successfully redistributed who is ultimately allowed — or required — to continue carrying them.